

Progress Review Report

Fulfillment of the work plan and objectives

The work completed so far has focused on two main parts: the theoretical design of the spillover model and the construction of the data pipeline that is needed to estimate the model.

At the beginning of the project, I reviewed academic work related to ETFs, comovement, arbitrage, liquidity and contagion. This was done to understand which could be the main mechanisms in which a shock to one stock could affect other stocks inside the same ETF.

The main idea of the thesis is that a shock to one ETF constituent may be transmitted to other constituents through different channels. The first channel is the weight of the shocked stock in the ETF, because a larger stock may have a stronger effect on the basket. The second channel is liquidity, as stocks with lower liquidity may react differently to shocks. The third one is ETF mispricing compared with the price of the constituents, which may capture trading effects. The fourth channel is economic similarity between constituents, as companies in the same sector or industry may react to similar information. Finally, the model also studies whether negative shocks behave differently from positive shocks.

The theoretical work was then translated into an econometric model where the shock is measured using the cumulative abnormal return of the shocked stock. Then, the dependent variable is the abnormal return of the receiving stock. The main variables in the regression are the size of the shock, the weight of the shocked stock, the illiquidity of the receiving stock, the ETF mispricing, the similarity between the two companies and a negative-shock indicator.

After defining the theoretical model, I coded the data pipeline, in which I prepared the ETF holdings data in a preprocessing step for five ETFs: SPY, XME, XLE, IHE and XLV. The original data was in JSON format, with one file per day, and it was converted into one aggregated CSV file for each ETF. During this process, I removed assets that were not stocks, as cash positions, invalid symbols and duplicated rows. Once the holdings data was prepared, I built the price and returns dataset, for which I downloaded daily adjusted prices using the yfinance library. I also added scripts to improve data quality, that identify failed tickers, clean ticker formats, deal with ticker changes and handle cases related to mergers, acquisitions and delisted companies.

After preparing the price data, I computed daily log returns. In this step I also removed tickers with too little data and filled only very short gaps, to finally have a clean dataset that can be used to estimate abnormal returns.

Then, a benchmark for each ETF was selected, for example, SPY uses the S&P 500 index, XME uses XLB, IHE uses XLV, XLV uses the S&P 500 Health Care index and XLE, the Brent price.

In parallel, I built the extra variables needed for the model, downloading volume data to calculate the Amihud illiquidity ratio. This variable measures price impact relative to trading volume, so it gives a better measure of liquidity than simple return volatility. I also downloaded GICS sector and industry data to measure how similar two companies are. Companies in the same industry are treated as more similar, companies in the same sector have medium similarity and companies in different sectors have lower similarity.

Then I implemented the spillover model, that defines a shock as a large cumulative abnormal return of a stock. For each stock, I estimate normal returns using a market model with a 120-trading-day estimation window and a 5-day gap before the event date. Then I calculate the cumulative abnormal return over the event window from t_0 to t_0+3 . If this value is far from the stock's normal historical behavior, the model identifies it as a shock.

For each shock, I build observations for the other stocks that were in the same ETF on that date, with the goal of testing whether the shock to one stock affects the abnormal returns of the other stocks in the same ETF via a regression.

A validation version of the model has already been tested with XME and XLE, that are the ETFs with less data. This first test gave preliminary evidence that spillovers exist inside ETFs.

Adjustments made and justification for these changes

The main adjustment is the way shocks are defined. At the beginning, the idea was to use external corporate events, such as earnings news, CEO resignations, product problems and regulatory announcements. In the current version, I define shocks directly from abnormal stock returns. I decided to make this change because it could be applied in a consistent way across all stocks and all dates, whereas getting corporate events data for all the constituents of all the ETFs during the window of time was not possible. With the current approach, the model can detect important stock movements even when the exact news event is not manually classified.

I also changed the idea of the liquidity variable, the first version used return volatility as a proxy for illiquidity, but I replaced it with the Amihud illiquidity ratio, which is more directly related to liquidity because it uses both returns and trading volume.

Work plan for completing the master's thesis

The next step is to identify improvements to make to the econometric model and run the final version of the model with all five ETFs. I would also like to use a more complex model with a neural network to compare the results with the econometric model.

I would also like to check if the results change when using alternative benchmarks and also compare different time periods, such as before and after COVID, to see whether spillovers change under different market conditions.

The final stage will be to write the full thesis, revise the results and include the feedback from my supervisors.